**表 1:** 消融实验主指标

| 层 | 能力 | AUROC | AUPRC | Cohen_d | Brier↓ | ECE↓ |
|---|---|---|---|---|---|---|
| L0 | Rules + CUSUM | **0.5075** | 0.9994 | 0.1728 | 0.1757 | 0.0647 |
| L1 | + PBC Detection | **0.9717** | 0.9939 | 5.1805 | 0.0692 | 0.1610 |
| L2 | + Evidence + Identity | **0.9717** | 0.9936 | 6.6620 | 0.0393 | 0.1023 |
| L3 | + Observer (Full System) | **0.9851** | 0.9967 | 7.0363 | 0.0372 | 0.1053 |